

Ex.No: 12

OBSERVATION:(Simulation)



INSTRUCTIONS

Select the clustering algorithm:

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Fig 1: Selecting the clustering algorithm (Agglomerative Clustering)



CHOOSE INSTRUCTIONS

Select the linkage type:

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Fig 2: Selecting the Linkage type (Single Linkage)

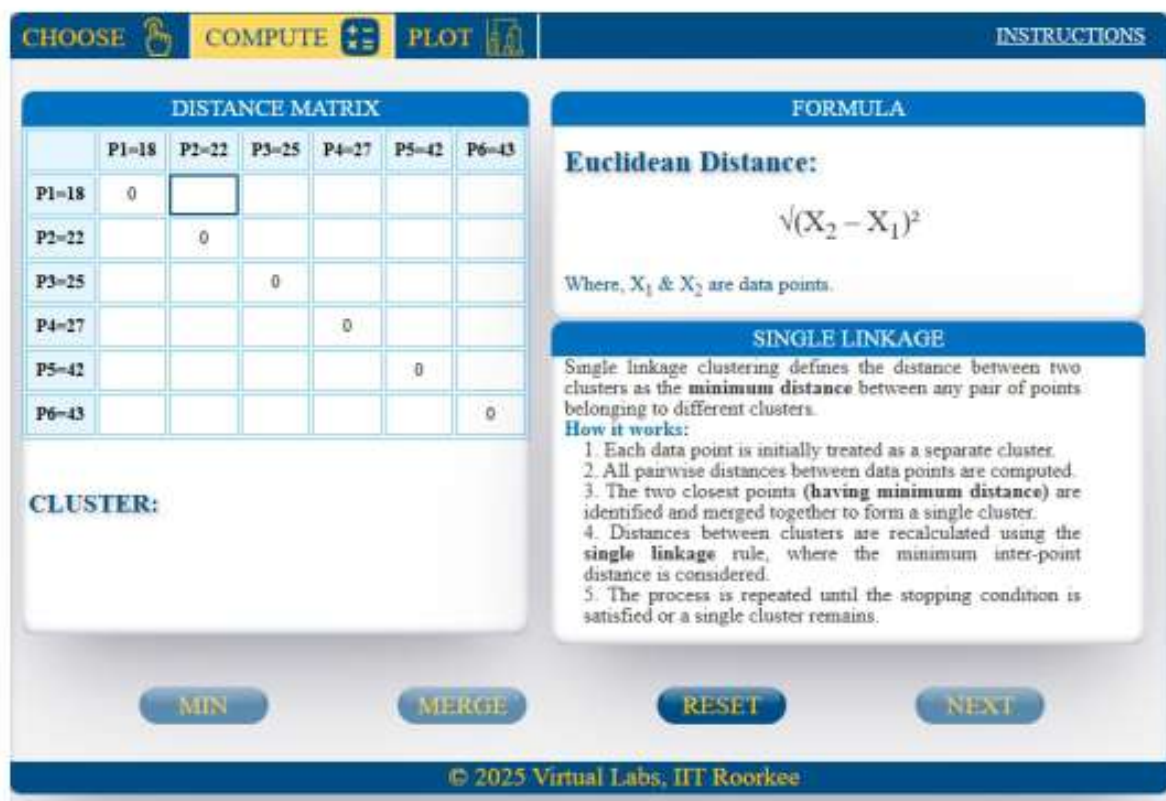


Fig 3: Initial distance matrix and formula display

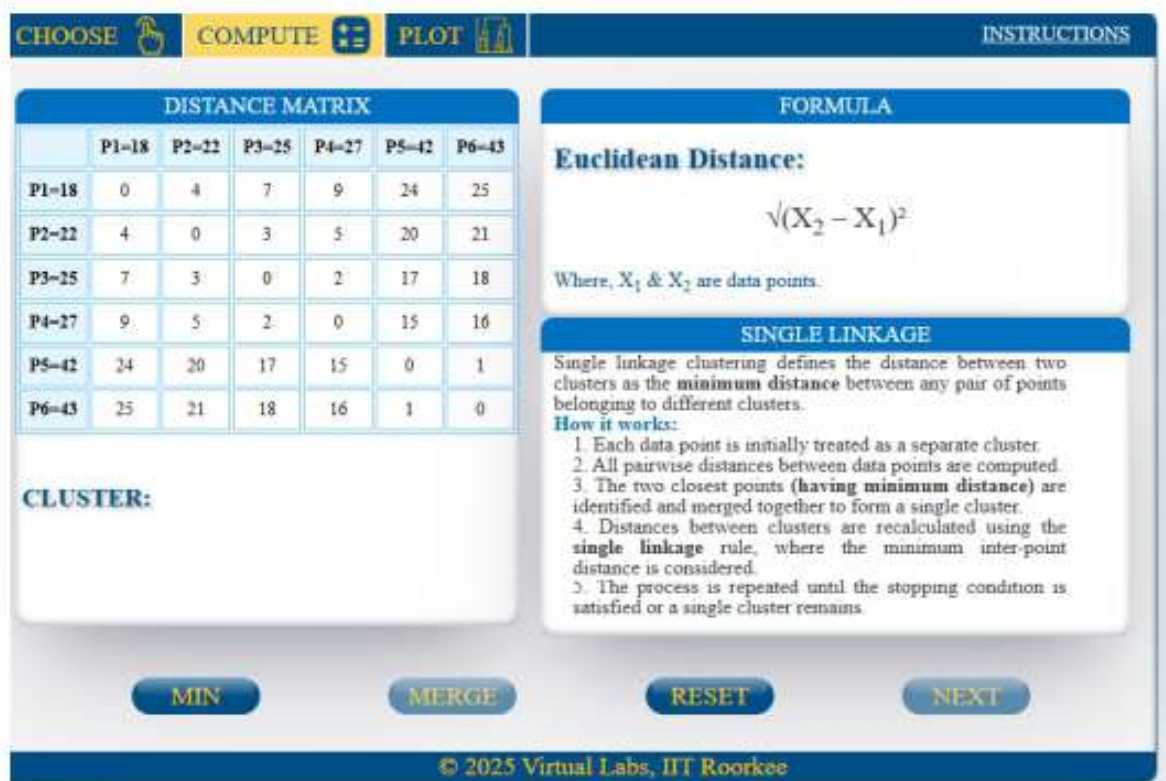


Fig 4: Calculated distance matrix for all data points

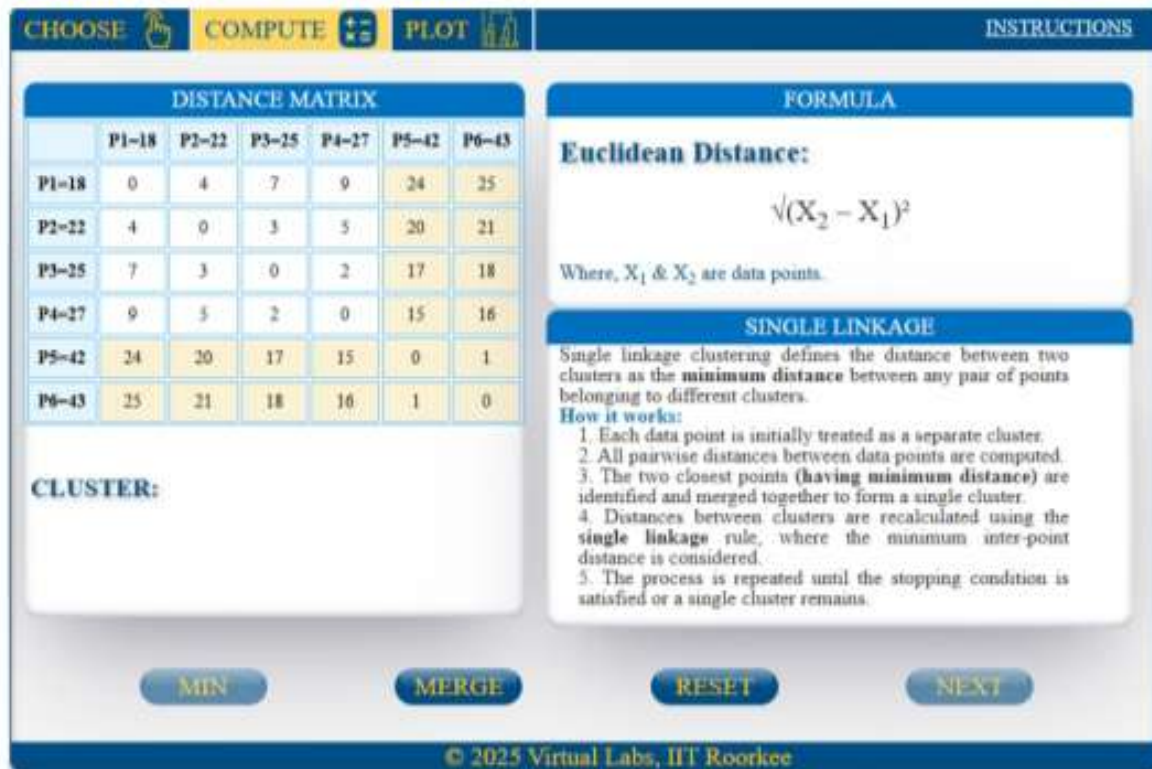


Fig 5: Finding the minimum distance in the distance matrix

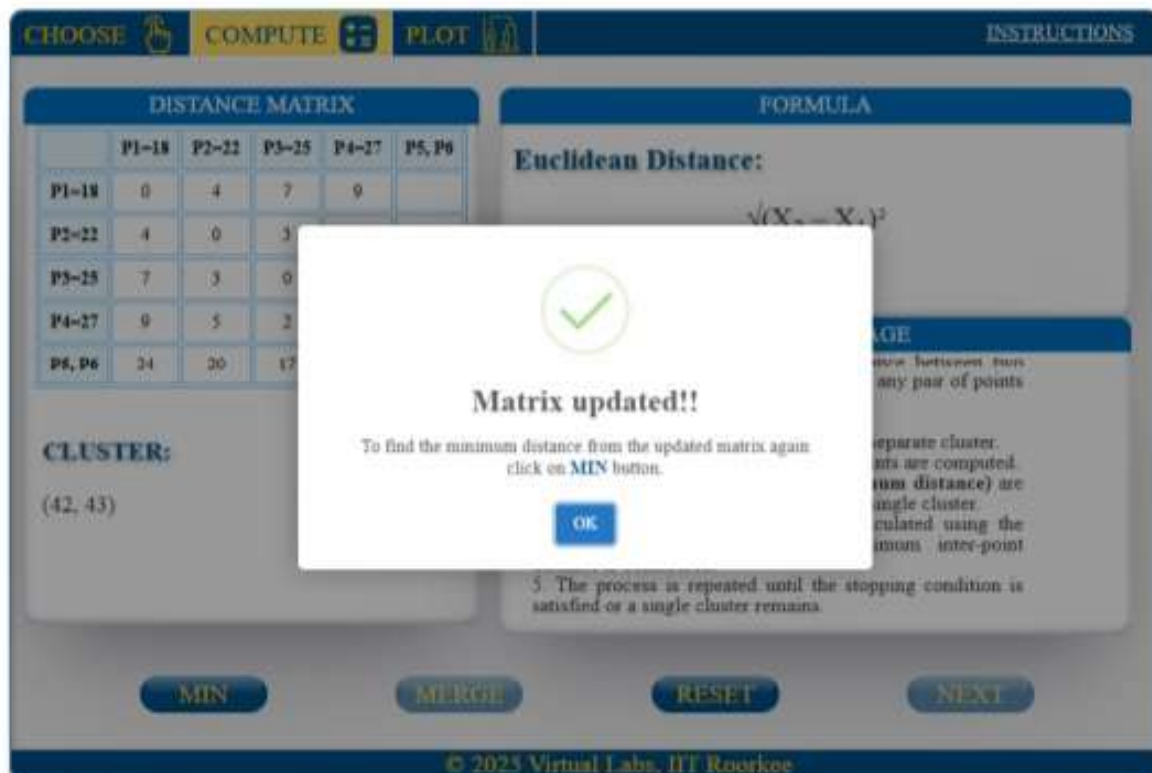


Fig 6: Popup confirmation after updating the distance matrix for cluster(42, 43)

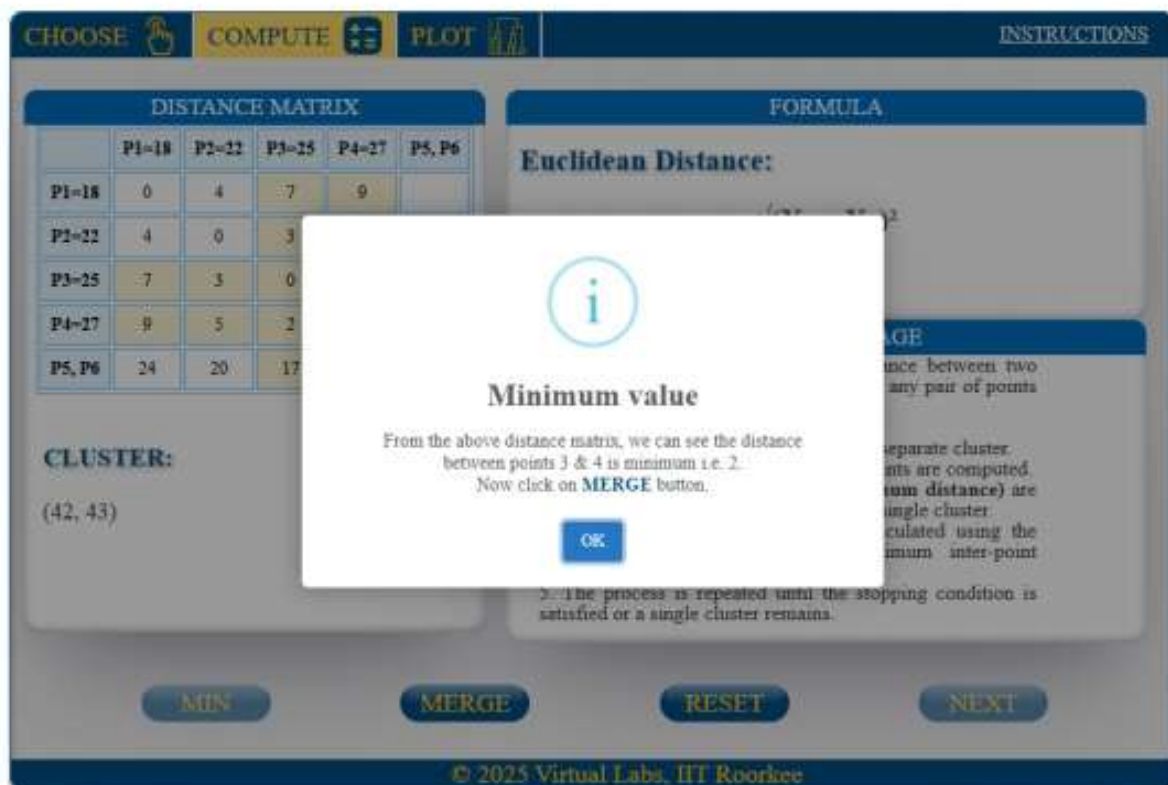


Fig 7: Highlighting minimum distance between points 3 and 4

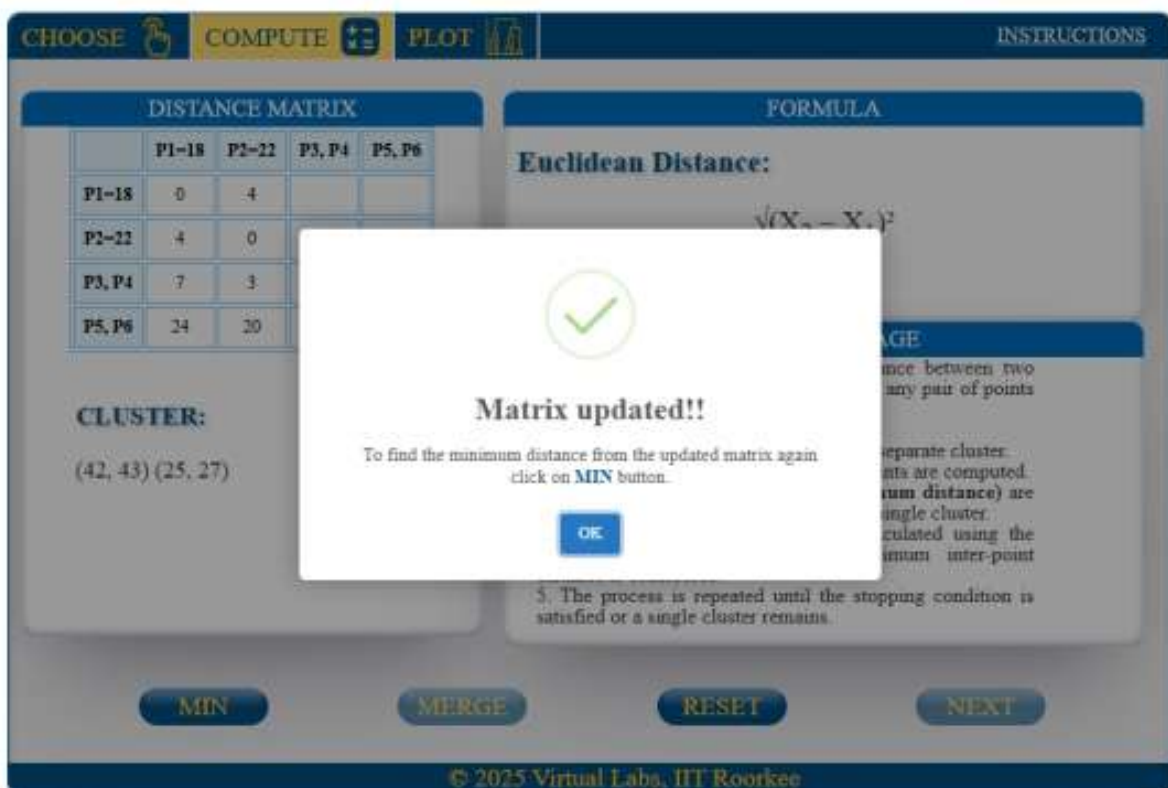


Fig 8: Popup confirmation after updating matrix for cluster (25, 27)

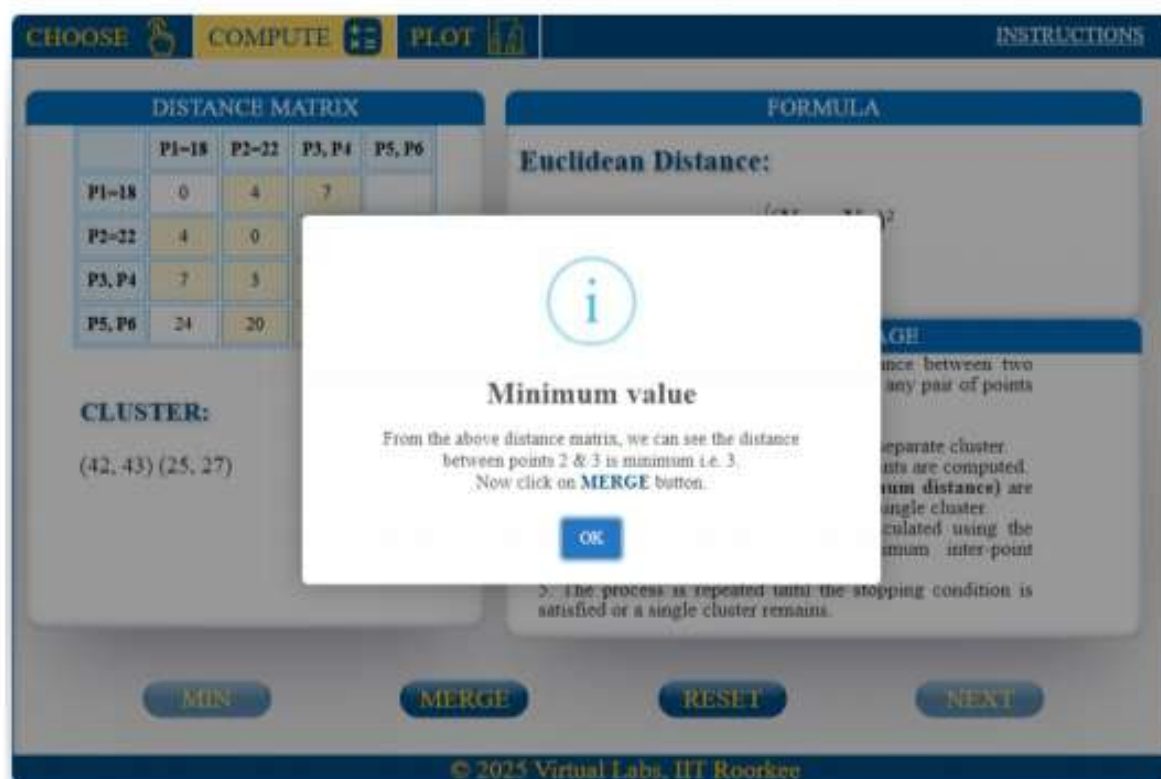


Fig 9: Highlighting minimum distance between point 2 and cluster (25, 27)

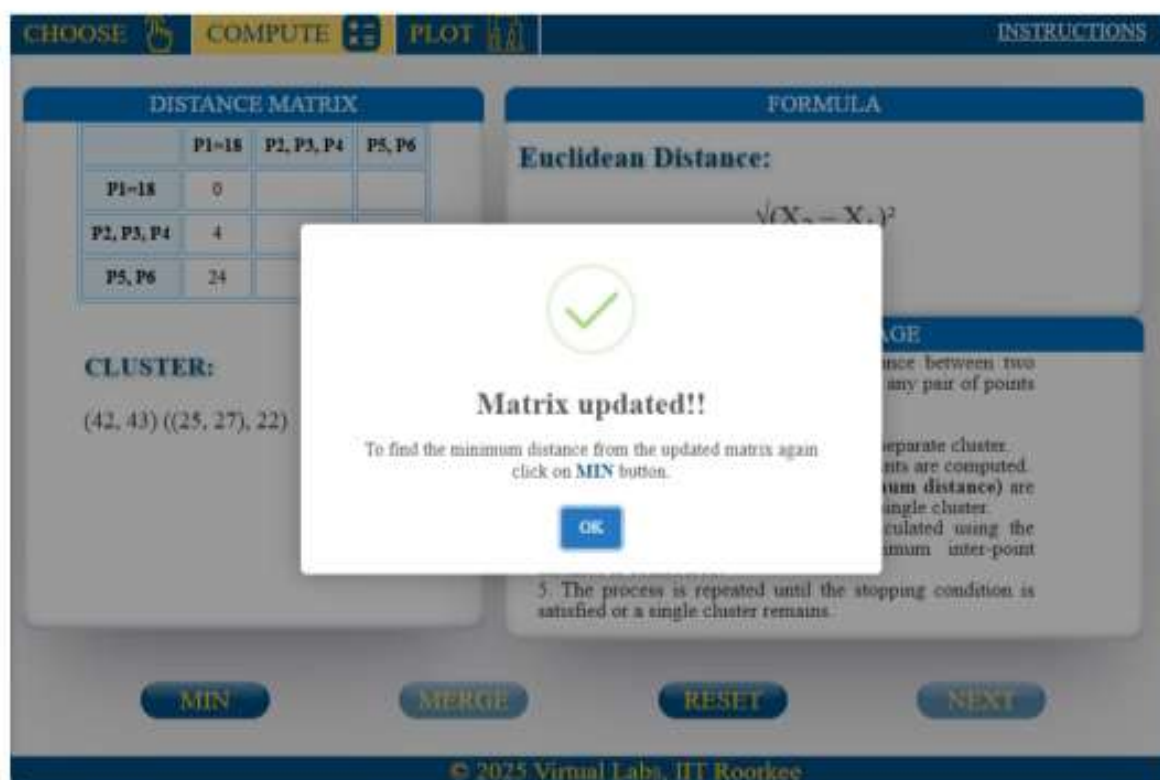


Fig 10: Popup confirmation after merging cluster ((25, 27), 22)

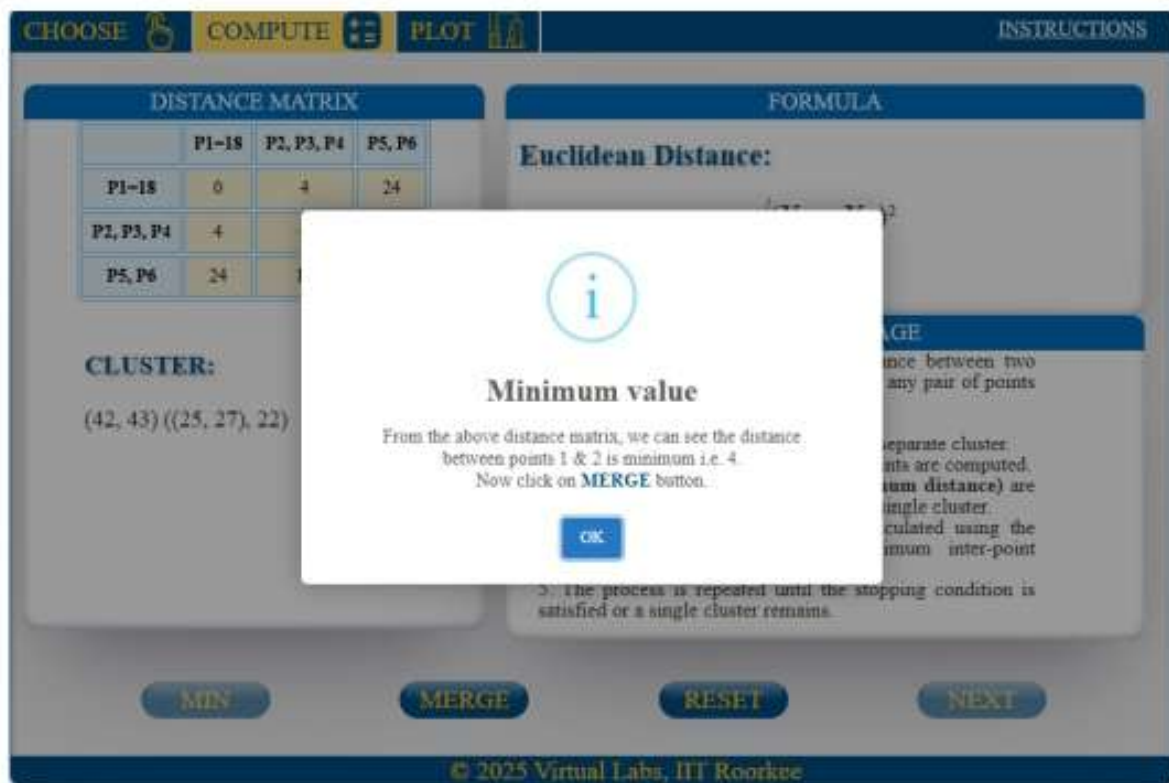


Fig 11: Highlighting minimum distance between point 1 and cluster((25, 27),22)

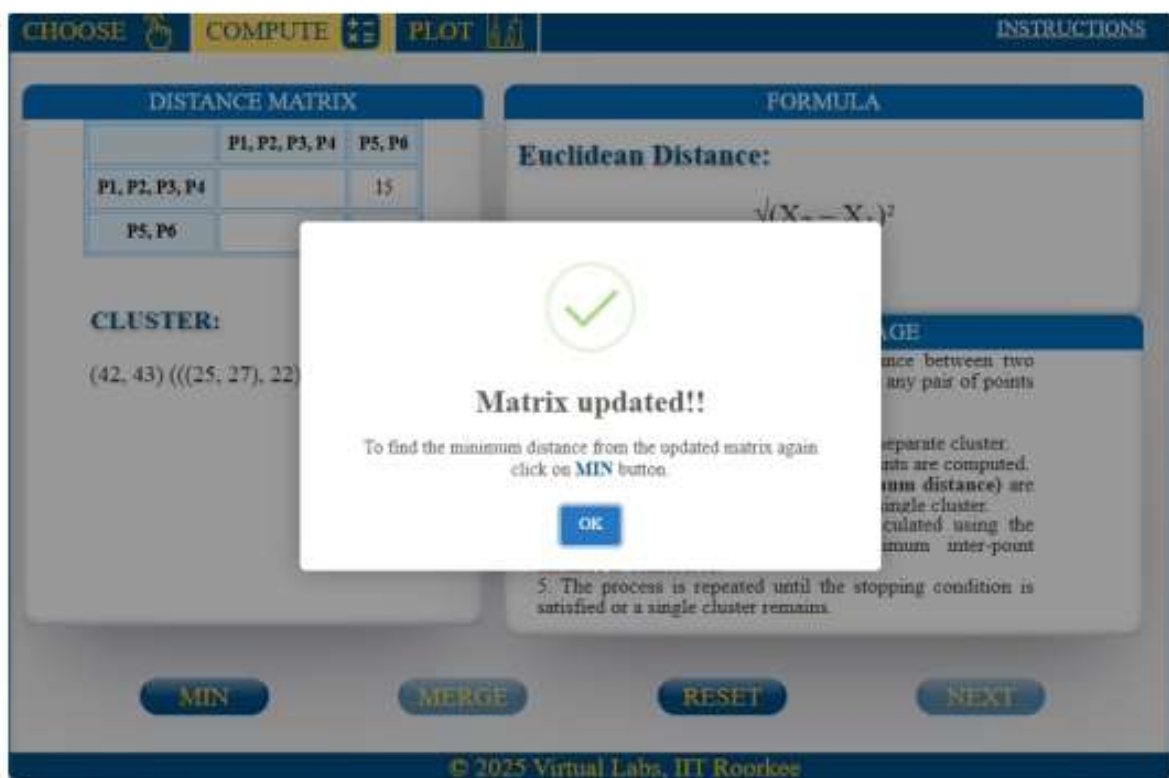


Fig 12: Popup confirmation after updating matrix for cluster(((25, 27), 22), 18)

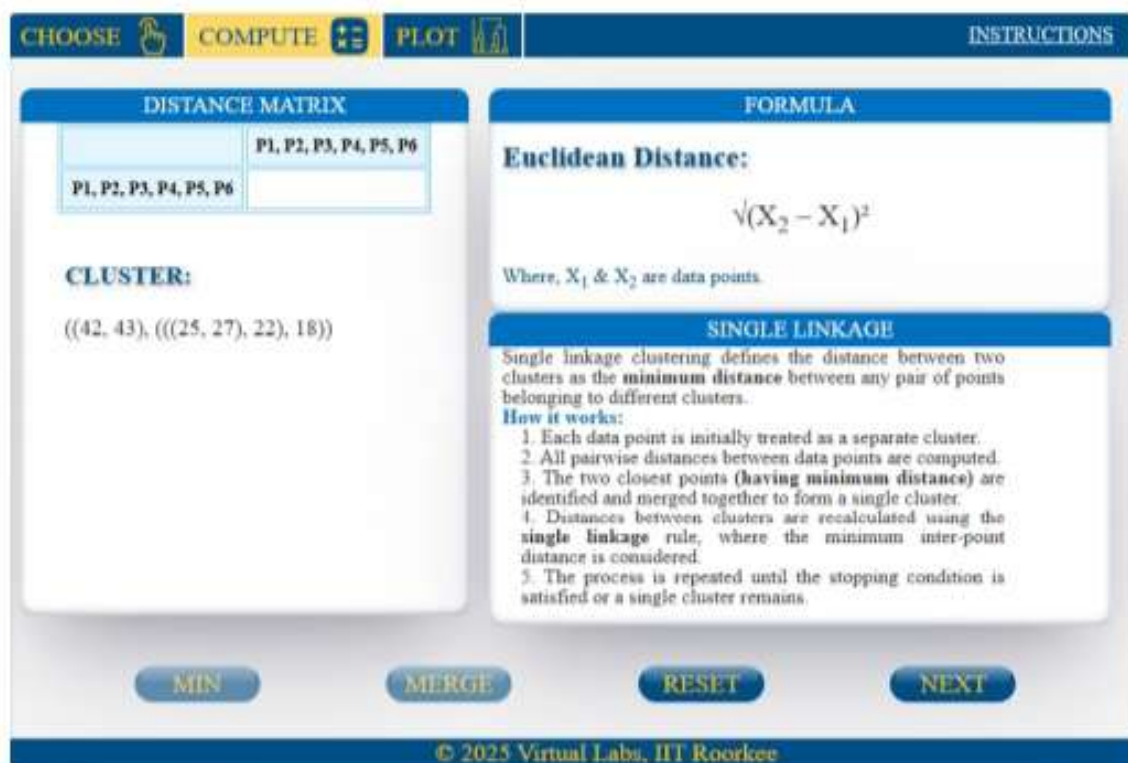


Fig 13: Final distance matrix calculation before total cluster merge



Fig 14: Prompt message asking to plot the dendrogram

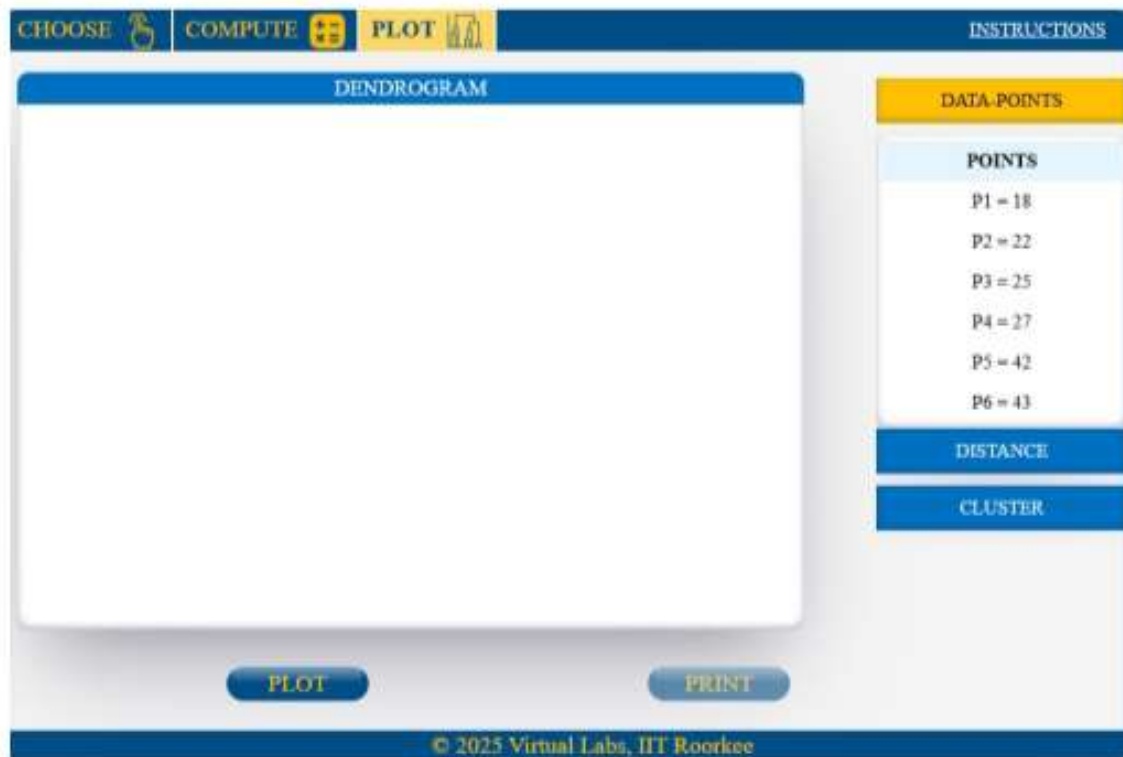


Fig 15: Viewing selected data points List



Fig 16: Viewing cluster distance values table



Fig 17: Viewing final merged cluster output string

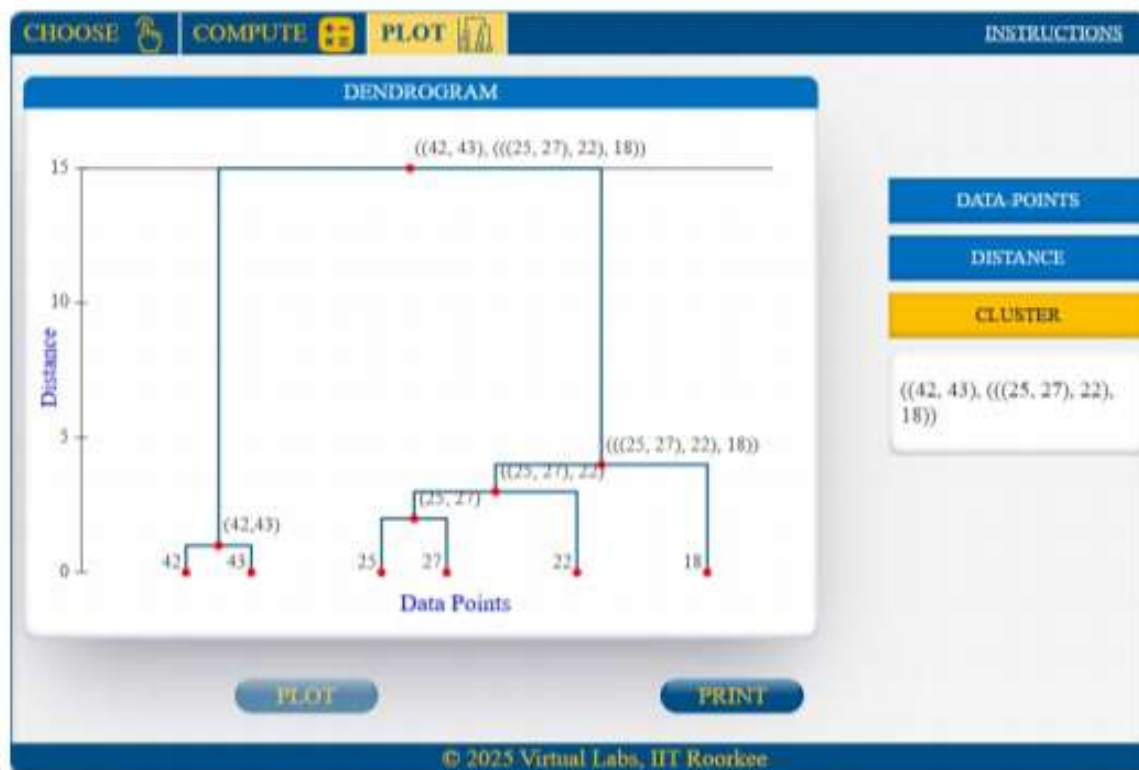


Fig 18: Final plotted dendrogram showing full hierarchical cluster tree

Simulation link: <https://ml-iitr.vlabs.ac.in/exp/hierarchical-clustering/simulation.html>